

MPL Assignment-2

03/05

Q1) A = PWA is a web applications that combines both web & mobile apps to deliver a seamless, PWA work offline, load quickly & provide app like experience

- Platform independence
- Improved performance
- Offline functionality
- No app store dependencies

⇒ Key characteristics :-

- (i) Installation: Installed from browser, traditional mobile apps are downloaded from mobile app
- (ii) Platform dependencies
- (iii) Offline support
- (iv) Updates
- (v) Performance
- (vi) It's an approach that ensures web pages adapt to different screen sizes & orientations using flexible grids
- Eliminates need for different devices and enhance usability

Feature	Responsive	Fluid	Adaptive
Definition	does CSS media queries to adjust	uses % for elements	uses pre defined layout
Flexibility	Highly flexible	completely flexible	Fixed & specific breakpoints
Performance	efficient but more adjustment	smooth	cause shifts
Best use case	website & PWAs for all screens	App requires scaling	pre defined layout

Q3) (i) Lifecycle Phases:

(ii) Registration:

```

→ if ('service worker' in navigator) {
    navigator.serviceWorker.register('sw.js')
    then (1) ⇒ console.log('Service registered')
}

```

(iii) Installation:

→ Occurs when the service worker is first downloaded

Eg: self.add EventListener ('install', event => {
 event.wait until C
 return . cache still
});

(ii) Activation

→ Runs after installation & ensures old cache cleared if necessary

Eg: self.add EventListener ('activate', event => {
 event.wait until C
 cache Keys C, then C Keys => {
 return [Keys . filter = 'v']
 }
});

(iv) Fetching & Updates

→ The service worker intercepts network requests

Indexed DB is a Low-level NoSql database, in the browser that allows web pages to store & retrieve

Service workers are JavaScript that run in background separate from main browser thread, enabling features like:

(i) Background Synchronization

(ii) Push Notifications

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(iii) offline functionality

iv) Chaching and data management

Advantages of using Indexed DB with service workers

1. offline - First functionality : Store and access data from Indexed DB when offline
2. Large storage Capacity : Store large number of structured data efficiently
3. Asynchronous API : Improves performance and avoids blocking the main thread
4. Persistent storage : Unlike Local storage, Indexed DB supports complex data types and is non-volatile